

that branch leads to a better score for us than do the possibilities growing from the right branch. This style of reasoning is the essence of how our AI will decide on the best move. It will build a game tree, check the outcomes at all terminal nodes of the game tree, and then use simple recursive reasoning to decide what move to make, in light of the possible futures that decision will open up.

You probably noticed that the game tree in [Figure 10-3](#) is woefully incomplete. It appears that there are only two possible moves (the left branch and the right branch), and that after each of those possible moves, our opponent has only two possible moves. Of course, this is incorrect; there are many choices available to both players. Remember that they can connect any two adjacent points in the lattice. The true game tree representing this moment in our game would have many branches, one for each possible move for each player. This is true at every level of the tree: not only do I have many moves to choose from, but so does my opponent, and each of those moves will have its own branch at every point in the tree where it's playable. Only near the end of the game, when nearly all the line segments have already been drawn, will the number of possible moves shrink to two and one. We didn't draw every branch of the game tree in [Figure 10-3](#), because there's not enough space on the page—we only had space to include a couple of moves, just to illustrate the idea of the game tree and our thought process.

You can imagine a game tree extending to any possible depth—we should consider not only our move and the opponent's response, but also our response to that response, and our opponent's response to that response, and so on as far as we care to continue the tree-building.

Building Our Tree

The game trees we're building here are different in important ways from the decision trees of Chapter 9. The most important difference is the goal: decision trees enable classifications and predictions based on characteristics, while game trees simply describe every possible future. Since the goal is different, so will be the way we build it. Remember that in Chapter 9 we had to select a variable and a split point to decide every branch in the tree. Here, knowing what branches will come next is easy, since there will be exactly one branch for every possible move. All we need